

Erik Jonsson School of Engineering and Computer Science

Department of Materials Science and Engineering

Materials Science and Engineering (BS)

Bachelor of Science in Materials Science and Engineering

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Orlando Auciello, Kyeongjae (KJ) Cho, Massimo V. Fischetti, Larry Hornbeck, Julia W. P. Hsu, Jiyoung Kim, Moon J. Kim, Manuel Quevedo-Lopez, Amy V. Walker, Robert M. Wallace

Associate Professors: Lev D. Gelb, William Vandenberghe, Walter E. Voit, Chadwin D. Young

Assistant Professors: Kyle McCall, Cormac Toher

Professors Emeriti: Yves J. Chabal, Bruce E. Gnade, Don Shaw

Assistant Professor of Instruction: Rebecca McClain

UT Dallas Affiliated Faculty: Kenneth J. Balkus Jr., Stuart Cogan, Matthew J. Goeckner, Fatemeh Hassanipour, Mark Lee, Hongbing Lu, Bing Lv, Anton V. Malko, Kenneth K. O, Lawrence J. Overzet, Shalini Prasad, Jason D. Slinker, Mihaela C. Stefan, Anvar A. Zakhidov

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

[RHET 1302](#) College Writing II¹

[ECS 2390](#) Professional and Technical Communication¹

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2414](#) Integral Calculus^{2, 3}

or [MATH 2417](#) Calculus I^{2, 3}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics⁴

[PHYS 2326](#) Electromagnetism and Waves⁴

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Option](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[CHEM 1311](#) General Chemistry I

[CHEM 1312](#) General Chemistry II

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 70-75 semester credit hours

In order to make satisfactory academic progress in Materials Science and Engineering, a student must meet all University requirements for course grades and GPA. In addition, no course may be taken until the student has obtained a grade of C or better in each of the required prerequisites.

Major Preparatory Courses: 20-25 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

[CHEM 1311](#) General Chemistry I

[CHEM 1112](#) General Chemistry Laboratory II

[CHEM 1312](#) General Chemistry II

[ECS 1100](#) Introduction to Engineering and Computer Science

[ECS 1210](#) Foundations in Engineering and Computer Science I

[ECS 1220](#) Foundations in Engineering and Computer Science II

[RHET 1302](#) College Writing II¹

[MSEN 1100](#) Materials in the Modern World

[MSEN 1300](#) Introduction to Materials Science and Engineering

[ENGR 2300](#) Linear Algebra for Engineers

Students may choose one of the following two sequences:

[MATH 2413](#) Differential Calculus^{2, 3}

[MATH 2414](#) Integral Calculus³

[MATH 2415](#) Calculus of Several Variables³

Or

[MATH 2417](#) Calculus I^{2, 3}

[MATH 2419](#) Calculus II³

Students may choose one of the following two sequences:

[PHYS 2325](#) Mechanics⁴

[PHYS 2125](#) Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves⁴

[PHYS 2126](#) Physics Laboratory II

Or

[PHYS 2421](#) Honors Physics I - Mechanics and Heat⁴

[PHYS 2422](#) Honors Physics II - Electromagnetism and Waves⁴

[PHYS 2126](#) Physics Laboratory II

Major Core Courses: 44 semester credit hours beyond Core Curriculum

[ECS 2390](#) Professional and Technical Communication¹

[MSEN 2111](#) Mechanical Properties of Materials Lab

[MSEN 2310](#) Mechanical Properties of Materials

[MSEN 3251](#) Materials Characterization and Thermodynamics Lab

[MSEN 3252](#) Metals and Ceramics Lab

[MSEN 3253](#) Electronic, Optical and Magnetic Properties of Materials Lab

[MSEN 3311](#) Thermodynamics of Materials

[MSEN 3312](#) Materials Characterization

[MSEN 3313](#) Physical Metallurgy

[MSEN 3314](#) Ceramics and Glasses

[MSEN 3316](#) Polymer Science and Engineering

[MSEN 3317](#) Kinetics and Phase Transformations

[MSEN 3320](#) Electronic, Optical, and Magnetic Properties of Materials

[MSEN 3324](#) Data Analysis and Statistics in Materials Science and Engineering

[MSEN 3325](#) Mathematical Methods in Materials Science and Engineering

[MSEN 4301](#) Materials Selection and Design

[MSEN 4402](#) Senior Design in Materials Science and Engineering

Technical Electives: 6 semester credit hours

Take 6 semester credit hours from the list below:

[MSEN 3303](#) Introduction to Semiconductor Technology and Manufacturing

[MSEN 3304](#) Introduction to Sustainable Energy Engineering

[MSEN 4305](#) Sustainable Electricity Generation

[MSEN 4306](#) Batteries and Energy Storage Devices

[MSEN 3301](#) Introduction to Nanoscience and Nanotechnology

[MSEN 3302](#) Microscopy, Spectroscopy, and Nanotech Instrumentation

[MSEN 3V11](#) Topics in Materials Science and Engineering

[MSEN 4391](#) Technology of Plasma

[MSEN 4V95](#) Undergraduate Research

Other upper-level courses may be taken towards the Technical Electives requirement with prior approval of the Department.

III. Elective Requirements: 3-8 semester credit hours

Free Electives: 3-8 semester credit hours

Both lower- and upper-division courses may count as free electives.

Some students may need to complete certain 1000 and/or 2000 mathematics courses (e.g., [MATH 1314](#) and [MATH 2312](#)) prior to taking [MATH 2413](#) or [MATH 2417](#). Some technical electives, such as [MSEN 4391](#), may require the completion of additional prerequisites.

Degree programs in the Erik Jonsson School of Engineering and Computer Science are governed by various accreditation boards that place restrictions on classes used to meet the curricular requirements of degrees they certify. For this reason, not all classes offered by the University can be used to meet elective requirements. Please check with your academic advisor before enrolling in classes you hope to use as free electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit

Minors

Minor in Nanoscience and Technology

18 semester credit hours

Goals for the Minor in Nanoscience and Technology

As the field of nanotechnology develops further, particularly in the north Texas region, The University of Texas at Dallas has an important role to play in the education of knowledge workers for the industry. The Minor in Nanoscience and Technology offered by the Department of Materials Science and Technology provides a means for undergraduate students to familiarize themselves with the concepts and principles of nanotechnology.

This minor has been designed around three core MSEN designated courses, the content of which is exclusively related to Nanoscience and Nanotechnology. The remaining nine semester credit hours of courses may be chosen from the list of courses below.

The contents of the courses [BIOL 4461](#), [CHEM 3322](#), and [PHYS 4301](#) are similar enough that only one of these three courses may count toward the minor. In addition, several lower-division electives have been included to provide streamlined access to the available upper-division electives.

Since the three core courses are all upper-division electives, only one of the remaining nine semester credit hours must be an upper-division course, although students may choose to challenge themselves by pursuing all nine semester credit hours as upper-division electives.

Educational Objectives for the Minor in Nanoscience and Technology On completion of the Minor program, students will:

- Have a comprehensive general education background
- Have a working knowledge of nanotechnology and nanoscience principles and industry applications
- Be able to apply key concepts in materials science, chemistry, physics, biology, and engineering to the field of nanotechnology
- Understand the societal and technology issues that may impede the adoption of nanotechnology
- Have the ability to communicate effectively and work collaboratively
- Be able to become successful professionals and, if they desire, be able to pursue graduate study
- Be able to identify career paths and requisite knowledge and skills for career change towards nanotechnology

Requirements for the Minor in Nanoscience and Technology

A total of 18 semester credit hours are required, consisting of three core classes (9 semester credit hours) and 9 additional semester credit hours of electives.

I. Core Requirements: 9 semester credit hours

[MSEN 3301](#) Introduction to Nanoscience and Nanotechnology

[MSEN 3302](#) Microscopy, Spectroscopy, and Nanotech Instrumentation

[MSEN 4V95](#) Undergraduate Research (in Nanotechnology)

II. Elective Requirements: 9 semester credit hours

Students must complete at least nine semester credit hours chosen from the following courses. At least one of the courses must be upper-division (3000 or 4000):

MSEN-designated courses:

[MSEN 3310](#) Introduction to Materials Science

[MSEN 4391](#) Technology of Plasma

or [EE 4391](#) Technology of Plasma

[MSEN 4V95](#) Undergraduate Research (in Nanotechnology)

Lower-division courses:

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

[PHYS 2303](#) Contemporary Physics

[MECH 2320](#) Mechanics of Materials

Upper-division courses:

[PHYS 4352](#) Concepts of Modern Physics

[MECH 4360](#) Fundamentals of Nanostructured Materials

[MECH 4370](#) Principles of MEMS

[EE 3310](#) Electronic Devices

[EE 3311](#) Electronic Circuits

[CHEM 4335](#) Polymer Chemistry

[CHEM 3472](#) Instrumental Analysis

[CHEM 4473](#) Physical Measurements Laboratory

[CHEM 3321](#) Physical Chemistry I

[CHEM 4355](#) Computational Modeling

Only one of the following courses may be used to count toward the Minor:

[BIOL 4461](#) Biophysical Chemistry

[CHEM 3322](#) Physical Chemistry II

[PHYS 4301](#) Quantum Mechanics I

Certificates

Certificate in Nanoscience and Nanotechnology

9 semester credit hours

A Certificate in Nanoscience and Nanotechnology is offered by the Department of Materials Science and Engineering. It can be obtained by completing the following (as well as any required prerequisites):

Required courses:

[MSEN 3301](#) Introduction to Nanoscience and Nanotechnology

[MSEN 3302](#) Microscopy, Spectroscopy, and Nanotech Instrumentation

And choose one of the following:

[MSEN 3310](#) Introduction to Materials Science

[MSEN 4391](#) Technology of Plasma

[MSEN 4V95](#) Undergraduate Research (in Nanotechnology) (3 semester credit hours or more)

Any upper level class that satisfies the [Minor in Nanoscience and Technology](#)

Certificate in Semiconductor Technology

9 semester credit hours

A Certificate in Semiconductor Technology is offered by the Department of Materials Science and Engineering. It can be obtained by completing the following (as well as any required prerequisites):

Required courses:

[MSEN 3303](#) Introduction to Semiconductor Technology and Manufacturing

[MSEN 3320](#) Electronic, Optical, and Magnetic Properties of Materials

Elective courses:

Choose 3 semester credit hours from the following:⁵

[MSEN 3301](#) Introduction to Nanoscience and Nanotechnology

[MSEN 3302](#) Microscopy Spectroscopy and Nanotech Instrumentation

[MSEN 3310](#) Introduction to Materials Science

[MSEN 3312](#) Materials Characterization

[MSEN 3340](#) Materials Processing

[EE 3310](#) Electronic Devices

[EE 4330](#) Integrated Circuit Technology

[EE 4391](#) Technology of Plasma

[EE 4371](#) Introduction to MEMS

or [MECH 4370](#) Principles of MEMS

[PHYS 4352](#) Concepts of Modern Physics

[PHYS 4371](#) Solid State Physics

[MECH 2340](#) Circuits and Applied Electronics

[MECH 3320](#) Heat Transfer

[MECH 3360](#) Materials Science and Engineering

[MECH 3370](#) Applied Thermodynamics

5. Each course has a set of prerequisites that will need to be met prior to taking the course.

Certificate in Sustainable Energy Engineering

9 semester credit hours

A Certificate in Sustainable Energy Engineering is offered by the Department of Materials Science and Engineering. It can be obtained by completing the following (as well as any required prerequisites):

Required courses:

[MSEN 3304](#) Introduction to Sustainable Energy Engineering

Choose 3 semester credit hours from one of the following courses:

[MSEN 4305](#) Sustainable Electricity Generation⁶

[MSEN 4306](#) Batteries and Energy Storage Devices⁶

Elective courses:

Choose 3 semester credit hours from any of the following upper-level Electives:

[CHEM 4370](#) Carbon Capture and Sequestration

[CHEM 4381](#) Green Chemistry and Green Fuels

[EE 4362](#) Introduction to Energy Conversion

[EE 4363](#) Introduction to Power Electronics

[ENGY 3302](#) Managing Power and Renewable Energy Assets; Sustainability

[MECH 3370](#) Applied Thermodynamics

[MECH 4365](#) Energy Analytics

[MECH 4380](#) HVAC Systems

[MSEN 3301](#) Introduction to Nanoscience and Nanotechnology

[MSEN 3340](#) Materials Processing

[MSEN 4305](#) Sustainable Electricity Generation⁶

Other sustainable energy-related courses can be approved by petition.

6. MSEN 4305 and MSEN 4306 must both be taken to have one of them count towards the Elective Requirement.

1. Semester credit hours fulfill the communication component of the Core Curriculum.

2. Three semester credit hours of Calculus are counted under Mathematics Core, and one semester credit hour of Calculus is counted under Elective Requirements.

3. MATH 2417 and MATH 2419 requirements can be fulfilled by completing MATH 2413, MATH 2414, and MATH 2415, as is commonly done when enrollment prerequisites for MATH 2417 are not met. Students who take MATH 2413, MATH 2414, and MATH 2415 may request three additional semester credit hours of Mathematics be applied to fulfill Free Elective credit requirements and may petition that one additional credit hour of Mathematics be applied to fulfill Technical Elective credit requirements.

4. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted as Component Area Option Core.

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